

4月24日. 周四. 第19次课.

Topic: 定积分的应用.

[中文课本: 第七章. 积分的应用. P243 ~ P263 ;

Calculus: Chapter 6. Application of Integration. P427 ~ 453 ;
Chapter 8. Further Application of Integration. P543 ~ P584]

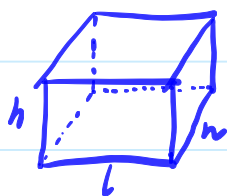
(2). 计算体积. (volume)

[中文课本: 7.2 体积的计算 P243 ~ P254 ;

Calculus: 6.2 Volumes P438 ~ P446.]

(1). 什么是体积?

e.g. 4.



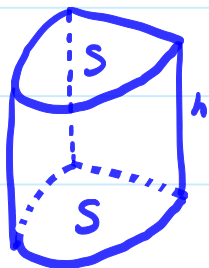
长方体 (Rectangular box):

$$V = h \cdot l \cdot w.$$



圆柱体 (Circular cylinder).

$$V = \underbrace{\pi \cdot r^2}_S \cdot h.$$



柱体 (cylinder)

$$V = S \cdot h$$

更一般的体积如何定义? 类似于面积、弧长的定义:
切成薄片, 近似求和.

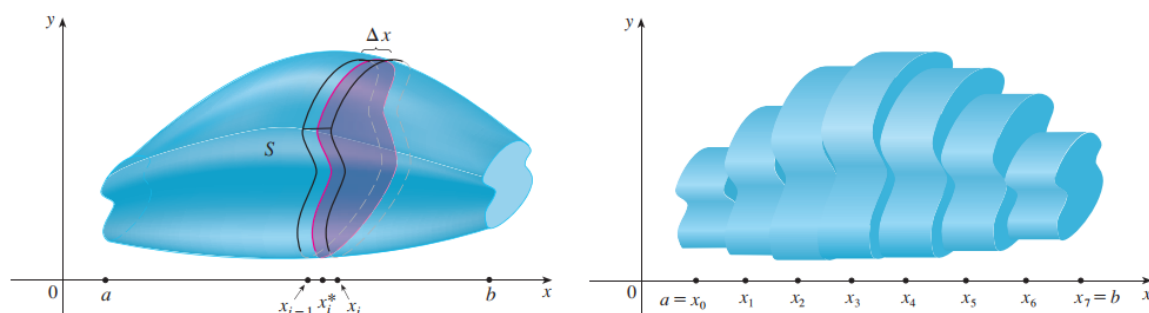


FIGURE 3

(2). 用定积分求体积.

体积公式: 设 S 是位于 $x=a$ 与 $x=b$ 之间的立体图形,

且 S 在 $x \in [a, b]$ 处被垂直于 x 轴的平面所截 面积为 $A(x)$,

A 是连续函数, 则 S 的体积为

$$V = \int_a^b A(x) \cdot dx$$

简单讲讲公式为什么对.

切成 n 个薄片 $P_0, P_1, P_2, \dots, P_{n-1}, P_n$. 记 $P_i(x_i, 0)$, $\Delta x_i = x_i - x_{i-1}$

用柱体近似的体积 $V_n = \sum_{i=1}^n A(z_i) \cdot \Delta x_i$. $z_i \in (x_{i-1}, x_i)$.

$$\therefore S \text{ 的体积} = \lim_{n \rightarrow \infty} V_n = \int_a^b A(x) dx$$

eg. 5. 证明: 半径为 r 的球 (sphere) 体积为 $V = \frac{4}{3}\pi r^3$. [Calculus: P439~P440].

(参照 Calculus 的图和过程).

$$\begin{aligned} V &= \int_{-r}^r \pi(r^2 - x^2) \cdot dx = \pi r^2 \cdot 2r - \left(\frac{1}{3}\pi \cdot x^3\right) \Big|_{-r}^r \\ &= 2\pi r^3 - \left(\frac{1}{3}\pi r^3 - \frac{1}{3}\pi(-r)^3\right) = \frac{4}{3}\pi r^3. \end{aligned}$$

(3). 旋转体 (solids of revolution) 的体积.

这一节中, 我们讨论一种特殊立体图形——旋转体的体积.

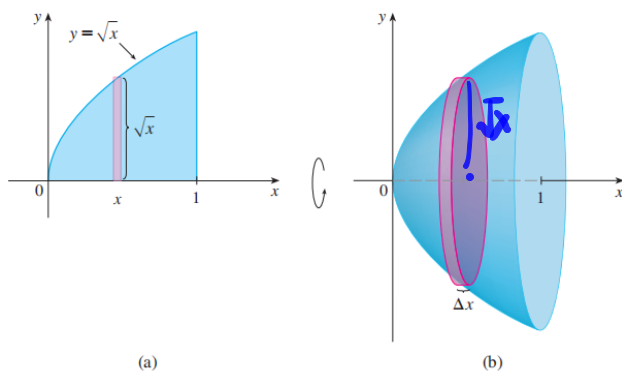
旋转体: 一个平面图形绕一条直线旋转一周后形成的立体图形.

通常我们会取这条直线为 x 轴, 并写出平面图形各边对应曲线的函数, 然后用上面讲的“切成薄片, 近似求和”的方式来求体积.

切成的薄片可以分成两类:

<1>. 盘形 (disk).

此时切出的薄片是圆形, 面积为 $\pi \cdot r^2$, 由此可确定 $A(x)$.



eg 6. 求在图中蓝色部分图形绕 x 轴旋转一周后生成的旋转体的体积.

(Calculus. P440, Ex 2).

$$A(x) = \pi \cdot (\sqrt{x})^2 = \pi x.$$

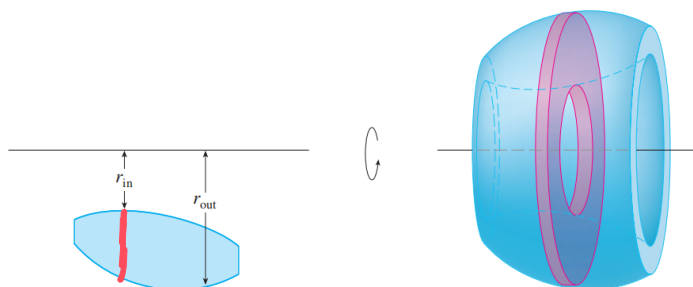
$$\therefore V = \int_0^1 A(x) \cdot dx = \int_0^1 \pi x \cdot dx = \left(\frac{\pi}{2} x^2 \right) \Big|_0^1 = \frac{\pi}{2}.$$

总结: 如果 $y = f(x) \geq 0, x \in [a, b]$, 那么这一段图形绕 x 轴旋转生成的旋转体体积 $V = \int_a^b \pi \cdot f^2(x) \cdot dx$.

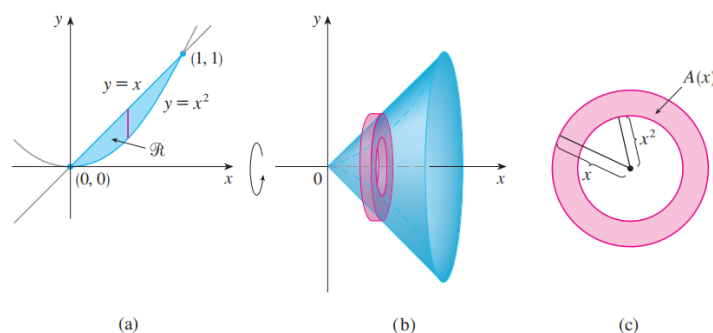
<2>. 垫圈形 (washer).

此时切出的薄片是圆环形.

面积为 $\pi \cdot (r_{\text{outer}})^2 - \pi \cdot (r_{\text{inner}})^2$.



e.g.7. 求出 $y=x$ 与 $y=x^2$ 所围成的图形绕 x 轴旋转一周后形成的旋转体的体积。
(Calculus. P442. EX 4.)

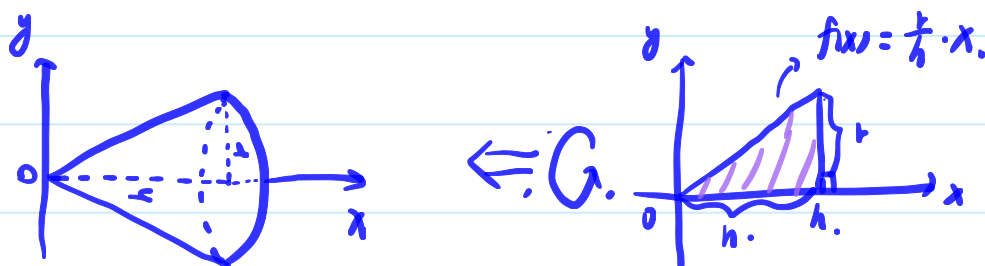


$$\therefore A(x) = \pi \cdot (x)^2 - \pi (x^2)^2 \\ = \pi \cdot x^2 - \pi \cdot x^4$$

$$\therefore V = \int_0^1 A(x) \cdot dx = \int_0^1 (\pi x^2 - \pi x^4) dx = \left(\frac{1}{3} \pi x^3 - \frac{1}{5} \pi x^5 \right) \Big|_0^1 \\ = \frac{\pi}{3} - \frac{\pi}{5} = \frac{2}{15} \pi.$$

Try it!

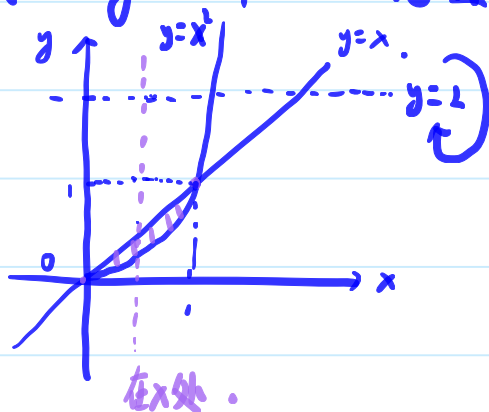
证明: 底面半径为 r , 高为 h 的圆锥 (cone) 体积为 $V = \frac{1}{3} \pi r^2 \cdot h$



$$A(x) = \pi \cdot f(x)^2 = \pi \cdot \frac{r^2}{h^2} \cdot x^2.$$

$$\therefore V_{\text{cone}} = \int_0^h A(x) dx = \int_0^h \frac{\pi r^2}{h^2} x^2 dx = \frac{\pi r^2}{h^2} \cdot \int_0^h x^2 dx \\ = \frac{\pi r^2}{h^2} \cdot \left(\frac{1}{3} x^3 \right) \Big|_0^h = \frac{\pi r^2}{h^2} \cdot \frac{h^3}{3} = \frac{1}{3} \pi r^2 \cdot h$$

e.g.7. 求 中图形绕直线 $y=2$ 旋转一周后形成的旋转体体积。
(Calculus. P442. EX 5)



$$r_{\text{inner}} = 2 - x$$

$$r_{\text{outer}} = 2 - x^2.$$

$$A(x) = \pi \cdot r_{\text{outer}}^2 - \pi \cdot r_{\text{inner}}^2 \\ = \pi \cdot (2 - x^2)^2 - \pi \cdot (2 - x)^2$$

$$A(x) = \pi \cdot (4 - 4x^2 + x^4) - \pi(4 - 4x + x^2)$$

$$= \pi \cdot (x^4 - 5x^2 + 4x)$$

$$\therefore V = \int_0^1 A(x) dx = \int_0^1 \pi \cdot (x^4 - 5x^2 + 4x) \cdot dx$$

$$= \pi \left(\frac{1}{5} \cdot x^5 - \frac{5}{3} \cdot x^3 + 2x^2 \right) \Big|_0^1$$

$$= \pi \cdot \left(\frac{1}{5} - \frac{5}{3} + 2 \right)$$

$$= \frac{8}{15} \pi.$$

(3) 计算旋转体的侧面积 (Area of a Surface of Revolution).

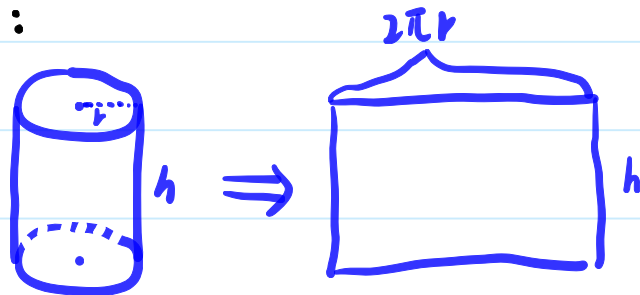
[中文课本: 7.4 旋转体的面积. (P258 ~ P263):

Calculus: 8.2 Area of a Surface of Revolution (P551 ~ P555)]

(1) 一些简单旋转体的侧面积:

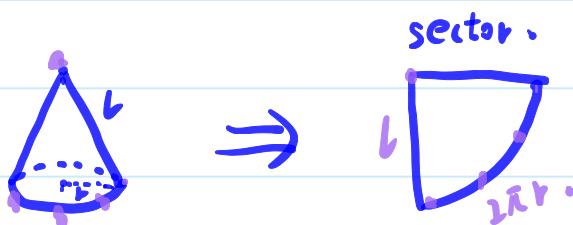
圆柱 (circular cylinder)

$$S = 2\pi r h$$

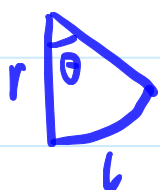


圆锥 (cone),

$$S = \frac{1}{2} \cdot l \cdot (2\pi r) = \pi \cdot r \cdot l$$



(如何求一个扇形 (sector) 的面积?)



idea: 它是 $\frac{\theta}{2\pi}$ 个圆 (circle).

$$\therefore l = \frac{\theta}{2\pi} \cdot (\text{圆的周长}) = \frac{\theta}{2\pi} \cdot 2\pi r = \theta \cdot r$$

$$S = \frac{\theta}{2\pi} (\text{圆的面积}) = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{\theta}{2} \cdot r^2$$

$$= \frac{1}{2} \cdot r \cdot l \quad (\text{很像 } S_{\Delta} = \frac{1}{2} a h \quad \triangle)$$

圆台 (frustum of a cone, or band?).

$$S = S_{\text{large sector}} - S_{\text{small sector}}$$

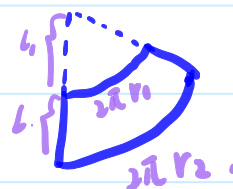
$$= (l_1 + l_2) \cdot 2\pi r_2$$

$$- l_1 \cdot 2\pi r_1$$

$$= 2\pi [l_1 r_2 + l_2 r_2 - l_1 r_1]$$

$$= 2\pi [r_1 l_1 + r_1 l_2 + l_2 r_2 - l_1 r_1]$$

$$= 2\pi l \cdot (r_1 + r_2) = \frac{1}{2} \cdot l \cdot [2\pi r_1 + 2\pi r_2] \quad (\text{类似于 } S_{\Delta} = \frac{1}{2} h(a+b) \quad \triangle)$$



$$\frac{r_1}{l_1} = \frac{r_2}{l_1 + l_2} = \cos \theta$$

$$\therefore r_1 \cdot (l_1 + l_2) = r_2 \cdot l_1$$

平均半径.

$$S = 2\pi \cdot \frac{r_1 + r_2}{2} \cdot r$$

(2). 用定积分求旋转体侧面积,

类似于体积的处理方式.

将旋转体切分成许多片.

每一片的侧面积用圆台近似.

第 i 片如图 (b), 由线段 $|P_{i-1}P_i|$

绕 x 轴转一周得到的圆台.

$$\text{它的侧面积: } S_i = 2\pi \cdot \frac{y_{i-1} + y_i}{2} \cdot |P_{i-1}P_i|$$

$$= \pi [f(x_{i-1}) + f(x_i)] \sqrt{(x_i - x_{i-1})^2 + [f(x_{i-1}) - f(x_i)]^2}$$

$$= \pi [f(x_{i-1}) + f(x_i)] |x_i - x_{i-1}| \cdot \sqrt{1 + [f'(x_i^*)]^2}$$

$$\therefore \text{总的侧面积} = \lim_{n \rightarrow \infty} \sum_{i=1}^n S_i$$

$$= \int_a^b \pi (f(x) + f(x)) dx \sqrt{1 + (f'(x))^2}$$

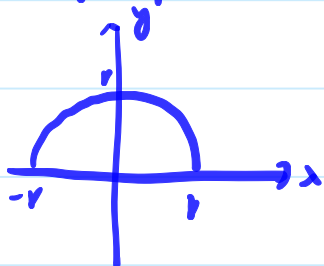
$$= \int_a^b 2\pi f(x) \sqrt{1 + (f'(x))^2} dx.$$

旋转体侧面积公式:

若 $f(x) \geq 0, x \in [a, b], f(x)$ 在 $[a, b]$ 上连续.

则 $f(x), x \in [a, b]$ 这一段绕 x 轴旋转一周所得侧面积 =

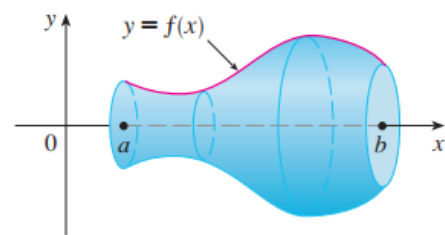
eg 8. 证明: 半径为 r 的球的表面积为 $4\pi r^2$.



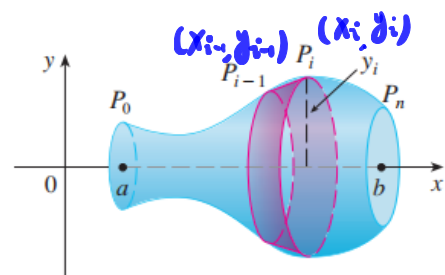
$$f(x) = \sqrt{r^2 - x^2} \quad \therefore f'(x) = \frac{-2x}{2\sqrt{r^2 - x^2}} = \frac{-x}{\sqrt{r^2 - x^2}}$$

$$\therefore S = \int_{-r}^r 2\pi \cdot \sqrt{r^2 - x^2} \cdot \sqrt{1 + \frac{x^2}{r^2 - x^2}} dx$$

$$= \int_{-r}^r 2\pi \cdot \sqrt{r^2 - x^2} \cdot \frac{r}{\sqrt{r^2 - x^2}} dx = \int_{-r}^r 2\pi r dx = 4\pi r^2.$$

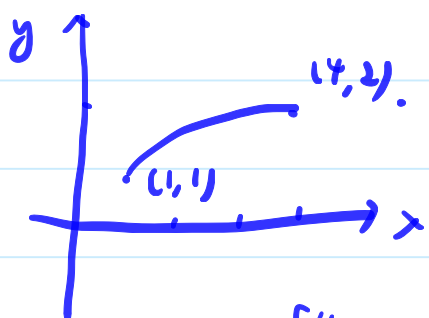


(a) Surface of revolution



(b) Approximating band

eg. 9. 求函数 $y = \sqrt{x}$ 从 $(1, 1)$ 到 $(4, 2)$ 这一段绕 x 轴旋转后形成的旋转体的侧面积.



$$f(x) = \sqrt{x}, \quad x \in [1, 4].$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$\therefore S = \int_1^4 2\pi f(x) \sqrt{1 + (f'(x))^2} dx$$

$$= \int_1^4 2\pi \cdot \sqrt{x} \cdot \sqrt{1 + \frac{1}{4x}} dx$$

$$= \int_1^4 2\pi \cdot \sqrt{x + \frac{1}{4}} dx$$

$$= \left[2\pi \cdot \left(x + \frac{1}{4}\right)^{\frac{3}{2}} \cdot \frac{2}{3} \right] \Big|_1^4$$